

Google Cloud SQL Database

Google Cloud SQL is a fully managed relational database service provided by **Google Cloud Platform (GCP)**. It allows users to deploy, manage, and scale popular relational databases such as **MySQL**, **PostgreSQL**, and **SQL Server** with minimal setup and maintenance. While “Google Cloud SQL Workspace” isn’t a specific product, the term may refer to using Cloud SQL within a broader GCP development or operations environment—such as Cloud Workstations, Cloud Shell, or application development environments.

Use Cases for Google Cloud SQL

- 1. Web and Mobile Applications**
Cloud SQL is commonly used to power backends for web and mobile applications, including content management systems (e.g., WordPress), e-commerce platforms, and customer relationship management (CRM) systems.
- 2. Business Applications**
Host traditional business applications such as ERP, HR, or finance systems that require relational database backends.
- 3. Data Warehousing and Analytics**
Cloud SQL can serve as a staging or operational database before transferring data to BigQuery or other data warehouses for advanced analytics.
- 4. SaaS Applications**
Supports multi-tenant architectures for Software-as-a-Service solutions, providing high availability and scalability.
- 5. Microservices Architectures**
Acts as a persistent data store for containerized or Kubernetes-based microservices.
- 6. Disaster Recovery and Backup**
Provides automated backups, cross-region replication, and point-in-time recovery to meet disaster recovery and compliance needs.
- 7. Development and Testing Environments**
Enables developers to quickly provision databases for development, staging, or continuous integration environments.

Benefits of Google Cloud SQL

Benefit	Description
Fully Managed	Automates infrastructure, patching, replication, and backups.
High Availability	Regional configurations provide automatic failover with minimal downtime.
Scalability	Easily scale instances and storage based on workload demands.

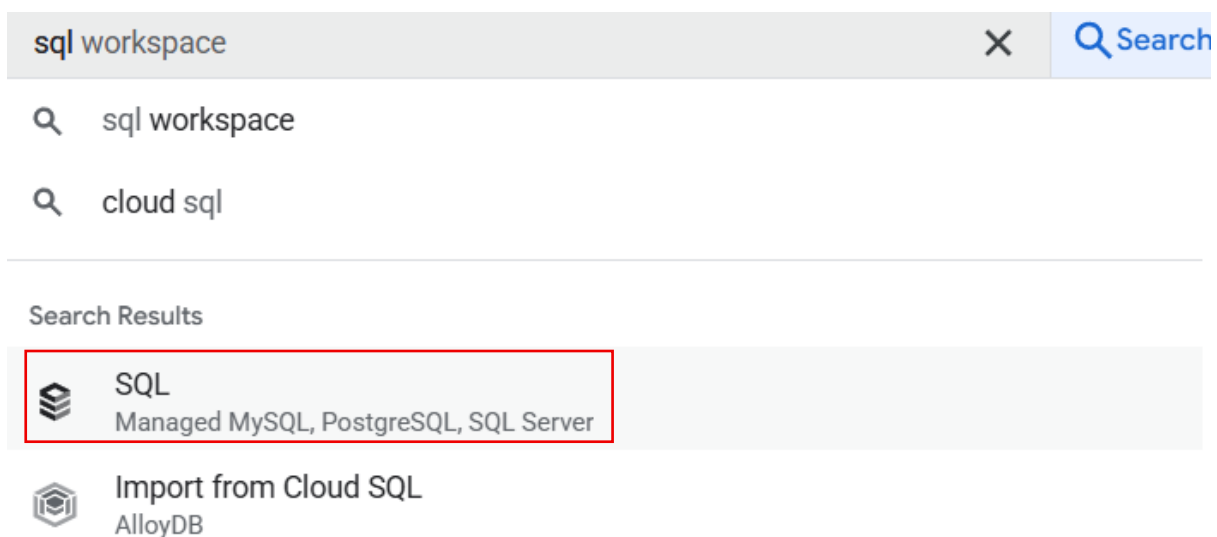
Benefit	Description
Security	Offers encryption at rest and in transit, IAM integration, private IP access, and VPC peering.
Automated Backups	Scheduled and on-demand backups with support for point-in-time recovery.
Performance Monitoring	Integrates with Cloud Monitoring and Query Insights for performance tuning and diagnostics.
Seamless Integration	Works natively with other GCP services such as BigQuery, Compute Engine, Dataflow, and App Engine.
Global Deployment	Deploy instances in multiple GCP regions to meet availability and latency requirements.

Complementary Tools for Workspace Integration

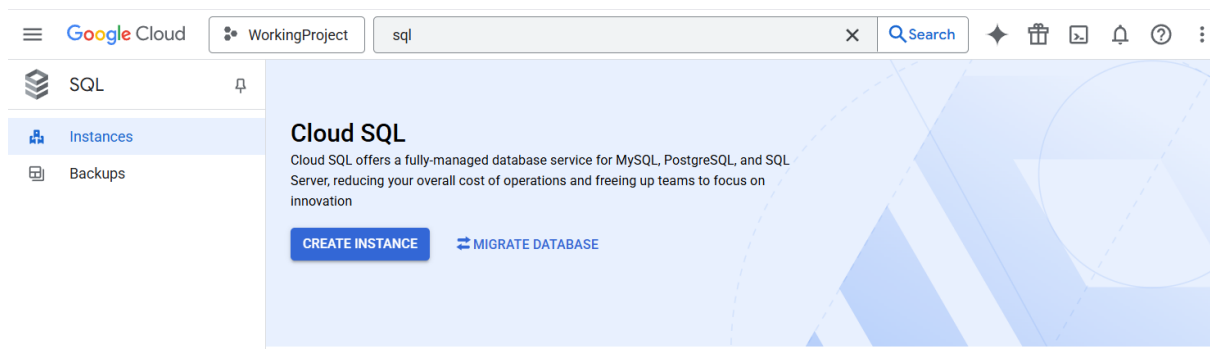
- **Cloud Shell or Cloud Workstations:** Secure, cloud-hosted development environments that connect easily to Cloud SQL.
- **Looker Studio or Dataform:** Analyze and transform data using SQL-based modeling tools.
- **Database Migration Service (DMS):** Migrate databases from on-premise or other cloud providers with minimal downtime.
- **Terraform or Google Cloud Deployment Manager:** Define and manage Cloud SQL infrastructure as code.

To begin with the Lab

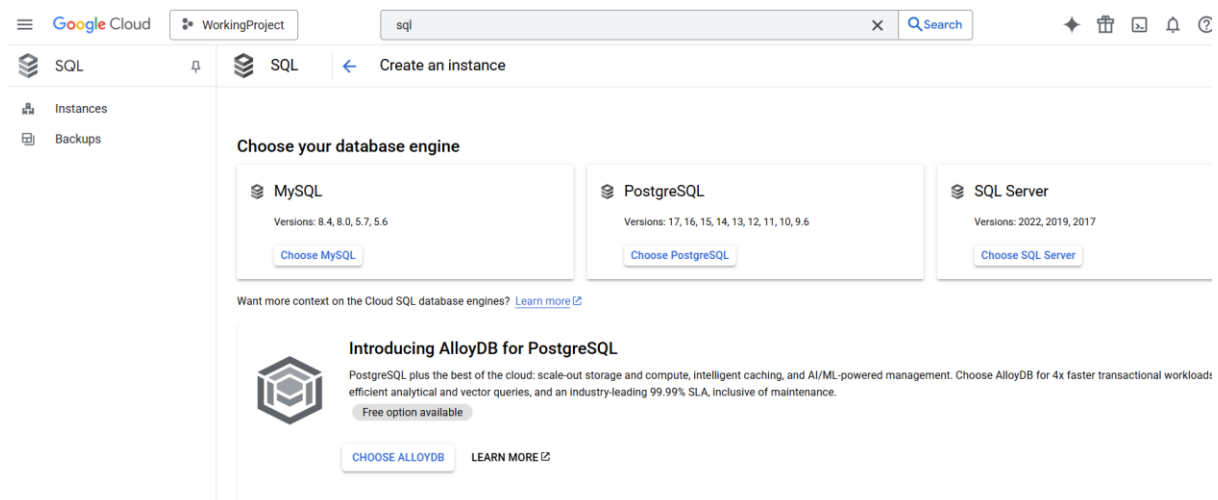
1. On your Google Cloud Console, search for SQL and choose the service accordingly.



2. Now from the dashboard of Cloud SQL click on Create Instance.



3. Then, you will see three database engines to choose from. You need to select MySQL. Click on choose and move forward.



4. Now, for the first option, you must choose the Enterprise edition for this lab.

Choose a Cloud SQL edition

A Cloud SQL edition determines foundational characteristics of your instance. Choose the best option for your price and performance needs. [Learn more](#) 

☐ Enterprise Plus

- 99.99% availability SLA
- Sub-second planned maintenance downtime
- Near-zero downtime instance scale-up
- Performance optimized machines
- Up to 35 days point-in-time recovery window
- Up to 3x higher read throughput with data cache
- Advanced disaster recovery with easy switchback
- Support for managed connection pool

☒ Enterprise

- 99.95% availability SLA
- Less than 60 seconds planned maintenance downtime
- General purpose machines
- Up to 7 days point-in-time recovery window

5. For the edition preset, choose Sandbox as it has the lowest number of CPUs and RAM, and it is much cheaper than the others, the best option for this lab.
6. Then you have to choose a database version; for now, leave it at the default.
7. In the end, just give a name and password to your instance.

Choose a preset for this edition. Presets can be customized later as needed.

Edition preset

Sandbox

▼

?

[COMPARE EDITION PRESETS](#)

Instance info

Database version *

MySQL 8.0

▼

▼ [SHOW MINOR VERSIONS](#)

Instance ID *

mysql-db

Use lowercase letters, numbers, and hyphens. Start with a letter.

Password *

●●●●●●●●



[GENERATE](#)

Set a password for the root user. [Learn more](#)

8. After that, scroll down and choose a region and availability zone as per your preference.

Choose region and zonal availability

For better performance, keep your data close to the services that need it. Region is permanent, while zone can be changed any time.

Region

us-central1 (Iowa)

▼

Zonal availability

- ☒ **Single zone**
In case of outage, no failover. Not recommended for production.
- ☐ **Multiple zones (Highly available)**
Automatic failover to another zone within your selected region. Recommended for production instances. Increases cost.

▼ [SPECIFY ZONES](#)

9. Now scroll down to connections and you will see that your DB instance has a public IP that will be attached to it while creation.
10. So, now we need to add an authorized network so that we can establish a connection with our DB instance.
11. For that, click on add a network, give a name to your network and then write the IP of your local machine.

Connections

Choose how you want your source to connect to this instance, then define which networks are authorized to connect. [Learn more](#) 

You can use the Cloud SQL Proxy for extra security with either option. [Learn more](#) 

Instance IP assignment

☐ Private IP

Assigns an internal, Google-hosted VPC IP address. Requires additional APIs and permissions. Can't be disabled once enabled. [Learn more](#) 

☒ Public IP

Assigns an external, internet-accessible IP address. Requires using an authorized network or the Cloud SQL Proxy to connect to this instance. [Learn more](#) 

Authorized networks

You can specify CIDR ranges to allow IP addresses in those ranges to access your instance. [Learn more](#) 

 my-network (106.219.152.63)	(Not saved) 
ADD A NETWORK	

12. In the end, just click on create instance, or you can explore the other options for your knowledge, but do not change anything, and click on create.
13. Now wait for your instance to get created, or what you can do is visit the link given below to download MySQL Workbench.

<https://www.mysql.com/products/workbench/>

14. Here you can see that our instance has been created, now click on it to go inside.

SQL

Instances

Backups

Instances

CREATE INSTANCE

MIGRATE DATABASE

SHOW INFO PANEL

LEARN

Starting Feb 1, 2025, all instances running community end-of-life versions of PostgreSQL and MySQL are under extended support. These instances will be charged for extended support from May 1, 2025. Upgrade your instances running end-of-life versions before May 1, 2025 to prevent additional charges. [Learn more](#)

VIEW AFFECTED INSTANCES

DISMISS

Filter

Enter property name or value

Status	Instance ID	Issues	Cloud SQL edition	Type	Public IP address	Private IP address	Instance connection name	Actions
☒	mysql-db		Enterprise	MySQL 8.0	34.57.248.19		still-kit-459403-e2us-	

15. As you can see here, we have some default databases that have been created by default. But we will create our own database, so click on Create DB.

SQL

Databases

Primary instance

Overview

Cloud SQL Studio

System insights

Query insights

Connections

Users

Databases

Backups

Replicas

Operations

All instances > mysql-db

mysql-db

MySQL 8.0

CREATE DATABASE

Name	Collation	Character set	Type
information_schema	utf8mb3_general_ci	utf8mb3	System
mysql	utf8mb3_general_ci	utf8mb3	System
performance_schema	utf8mb4_0900_ai_ci	utf8mb4	System
sys	utf8mb4_0900_ai_ci	utf8mb4	System

16. To create a Database, you just have to give a name and click on Create.

Create a database

Database Name *
mysql-db

Must follow the MySQL identifier rules. [Learn more](#)

Character set *
utf8mb4

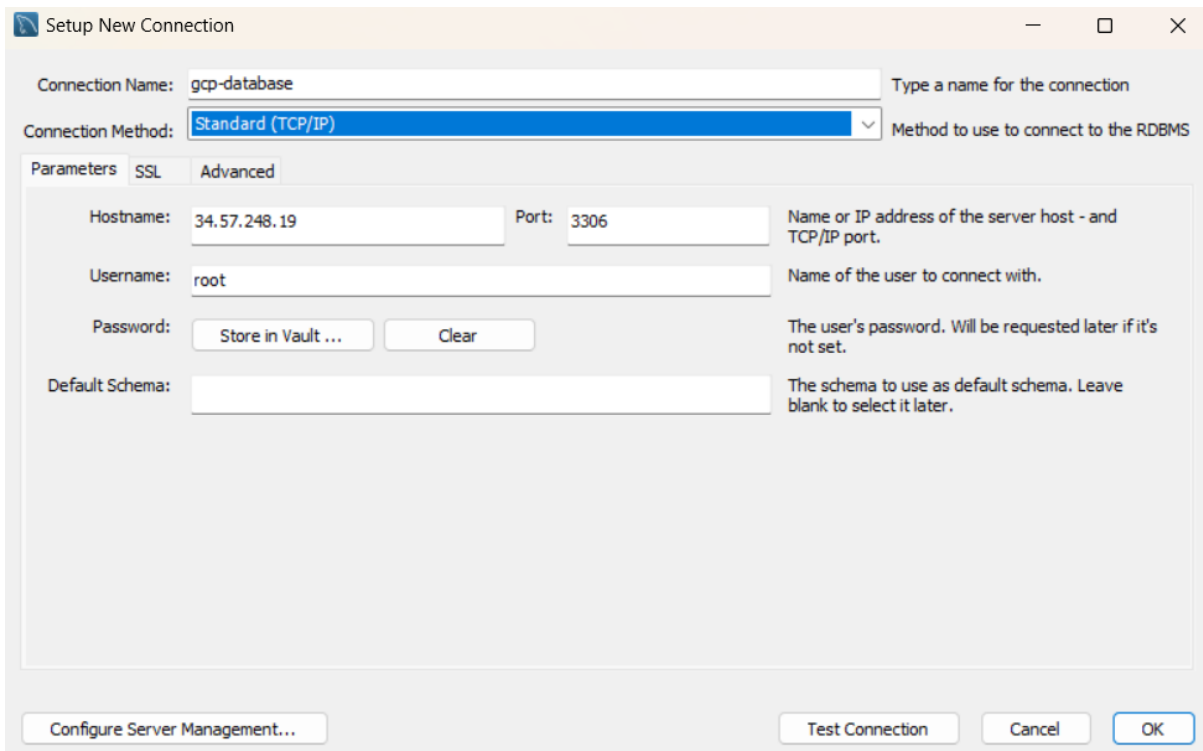
Can be changed later by executing an ALTER DATABASE query.

Collation
Default collation

Can be changed later by executing an ALTER DATABASE query.

CREATE **CANCEL**

17. After creating your Database, open the MySQL Workbench tool and create a new connection there.
18. Give a connection name, and then for the host name you have to give the public IP of your database instance.



The screenshot shows the 'Setup New Connection' dialog box in MySQL Workbench. The 'Connection Name' field is set to 'gcp-database'. The 'Connection Method' is set to 'Standard (TCP/IP)'. The 'Parameters' tab is selected, showing the following fields: 'Hostname' (34.57.248.19), 'Port' (3306), 'Username' (root), 'Password' (with 'Store in Vault ...' and 'Clear' buttons), and 'Default Schema' (empty). To the right of these fields are descriptive labels: 'Name or IP address of the server host - and TCP/IP port.', 'Name of the user to connect with.', 'The user's password. Will be requested later if it's not set.', and 'The schema to use as default schema. Leave blank to select it later.' At the bottom, there are buttons for 'Configure Server Management...', 'Test Connection', 'Cancel', and 'OK'.

19. To get the public IP address of your DB instance, you need to go back to your GCP console.
20. From the overview page of your instance, you can get the Public IP address. Copy it.

The screenshot shows the AWS Cloud SQL console interface. On the left, there's a sidebar with 'Primary instance' and a list of navigation items: Overview (selected), Cloud SQL Studio, System insights, Query insights, Connections, Users, and Databases. The main panel is titled 'Connect to this instance' and contains a table of connection parameters:

Connection name	still-kit-459403-e2:us-central1:mysql-db
Private IP connectivity	Disabled
Public IP connectivity	Enabled
Public IP address	34.57.248.19
Default TCP database port number	3306

21. Now you need to click on test connection first before actually connecting with your DB instance.
22. So, click on Test connection. Then you need to give the password that you defined while creating it. And click on test. You will see a window that says Connection successful.
23. Then click on OK and go inside your Database.

MySQL Workbench

Successfully made the MySQL connection

Information related to this connection:

Host: 34.57.248.19

Port: 3306

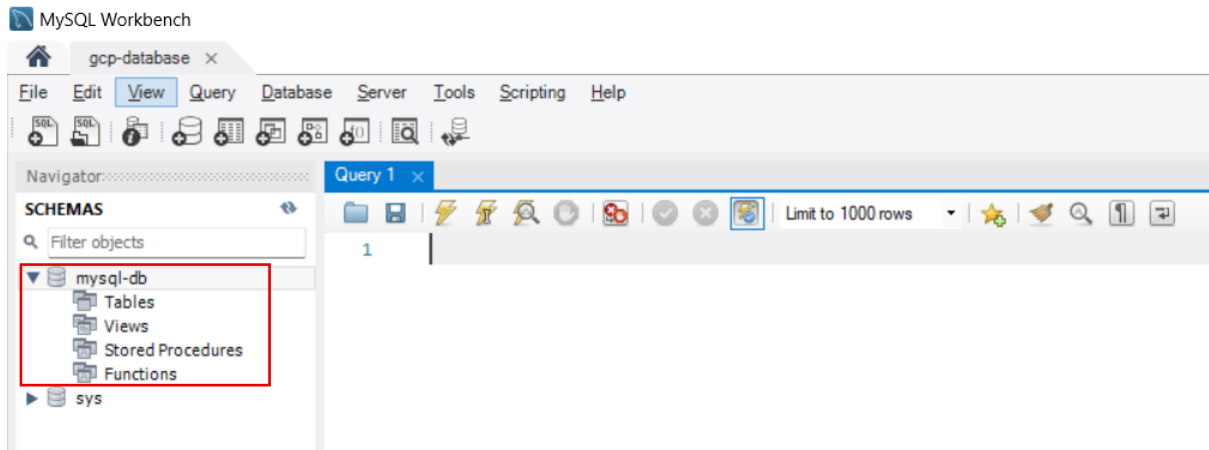
User: root

SSL: enabled with ECDHE-RSA-AES128-GCM-SHA256

A successful MySQL connection was made with the parameters defined for this connection.

OK

24. And here you can see your Database.



25. Now you can use it to run the basic SQL commands and have a basic understanding of your DB.

26. In the below snapshot, we used our database and then created a table, inserted some data init and then we displayed the data.

